

# Algebra I Foundations Cheatsheet

## Module 1

### Order of Operations

Parentheses  $\rightarrow$  Exponents  $\rightarrow$  Multiply/Divide  $\rightarrow$  Add/Subtract

Multiplication and division go left to right. Addition and subtraction go left to right.

### Signed Numbers

$$\begin{aligned} (+)(+) &= +, & (-)(-) &= +, & (+)(-) &= -, & (-)(+) &= - \\ a - b &= a + (-b) \end{aligned}$$

Add numbers with the same sign by adding absolute values and keeping the sign. Add numbers with different signs by subtracting absolute values and keeping the sign of the larger absolute value.

### Fractions

$$\begin{aligned} \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd}, & \frac{a}{b} - \frac{c}{d} &= \frac{ad - bc}{bd} \\ \frac{a}{b} \cdot \frac{c}{d} &= \frac{ac}{bd}, & \frac{a}{b} \div \frac{c}{d} &= \frac{a}{b} \cdot \frac{d}{c} \end{aligned}$$

Restrictions:

$$b \neq 0, \quad d \neq 0$$

### Evaluating Expressions

Substitute first, then simplify.

$$3x^2 - 2x \quad \text{at } x = -4$$

means

$$3(-4)^2 - 2(-4)$$

Use parentheses when substituting negative numbers.

### Like Terms

Like terms have the same variable part and the same exponents.

$$7x^2 - 3x^2 = 4x^2$$

$$5a + 2b - 3a = 2a + 2b$$

## Distributive Property

$$a(b + c) = ab + ac$$

$$-a(b - c) = -ab + ac$$

Common mistake:

$$3(x + 5) \neq 3x + 5$$

## Basic Exponents and Radicals

$$x^1 = x, \quad x^0 = 1 \quad (x \neq 0)$$

$$x^m \cdot x^n = x^{m+n}, \quad \frac{x^m}{x^n} = x^{m-n} \quad (x \neq 0)$$

$$(x^m)^n = x^{mn}$$

$$\sqrt{x^2} = |x|$$

## Algebraic Properties

$$a + b = b + a, \quad ab = ba$$

$$(a + b) + c = a + (b + c), \quad (ab)c = a(bc)$$

$$a(b + c) = ab + ac$$

## Quick Checks

- Combine like terms only when the variable parts match exactly.
- Keep fraction bars as grouping symbols.
- Use parentheses for negative substitutions.
- Never divide by zero.